When Access Should Change: Contextual Integrity, Encryption, and Accountable Transitions in a Religious Directory

YUXING YANG, University of Illinois Urbana-Champaign, USA

Religious directories combine household representation, volunteer roles, and continuing relationships. When a 17-year-old member takes over their settings, a support relationship ends, or a member objects to promotional email, the context of an information flow changes: some grounds for access end while others persist. Drawing on contextual integrity and on public policies of The Church of Jesus Christ of Latter-day Saints in a stipulated UK data-protection setting, we model each transition as separate decisions about management, receipt, restriction, and retention, and locate where exposure arises along the transition's timeline. In a synthetic evaluation of 46 authorization configurations, mechanism-level staleness is the most tractable source: a strict lease of $b + 1$ ticks meets a staleness budget of b ticks under the stated event ordering, and revision checks eliminate observed false grants at an availability cost. Controlling staleness leaves other exposures in place. A review-first no-contact policy, executed correctly, permits all 73 promotional-outreach probes that interim restriction blocks; a secondary path omitted from the probe set serves stale grants that primary-path evidence cannot see; and field-level encryption protects later values but cannot recall plaintext or key–ciphertext pairs already released. We therefore propose claim-scoped evidence records that bind a legally classified requirement to a declared query, path, and time scope, and that report coverage gaps alongside errors.

CCS Concepts: • **Security and privacy** → **Privacy protections**; **Access control**; • **Social and professional topics** → *Computing / technology policy*.

Additional Key Words and Phrases: contextual integrity, youth privacy, religious directories, dynamic authorization, encryption, data protection by design, accountability

1 Introduction

Consider S, a synthetic 17-year-old whose household representative manages contact settings. S requests control of those settings and narrower telephone sharing. Later, an assigned support relationship ends, while a separate event-email entitlement remains valid. S then asks to stop promotional email and eventually leaves the directory. What changes: the representative's management powers, a volunteer's reading permission, the organization's retention decision, or all three? What would demonstrate that a justified change was implemented?

Public policies of The Church of Jesus Christ of Latter-day Saints describe household settings, adult children's controls, and exceptions for organizational appointments (*callings*) and assigned support relationships (*ministering*) [1]. Its Privacy Notice discusses retention and rights requests [2]. The case is useful because these relationships coexist: household representation separates management from reading, callings and ministering separate dependent from independent grants, membership-linked contact data makes purpose and recipient matter, and departure separates storage from use. Backend transition behavior is unknown.

Contextual integrity (CI) holds that an information flow is appropriate when it conforms to the norms of its context, expressed through sender, recipient, subject, information type, and transmission principle [14]. Sending S's telephone number to a ward volunteer fits the norms of an assigned support relationship. When that relationship ends, when S takes over settings, or when S objects to outreach, the context changes and so do the governing norms. A transition is therefore more than revoking a credential. A representative can lose authority to change settings while a separately justified reading permission persists; a record can remain stored without remaining

Author's Contact Information: Yuxing Yang, University of Illinois Urbana-Champaign, Champaign, Illinois, USA, yuxingy2@illinois.edu.

available for outreach. Encryption can bind a new value to the new norms, but a value released under the old norms stays with its recipient.

We argue that the evidence needed to assess a transition depends on *where along its timeline exposure can arise*: before a request is classified, within an enforcement window, on paths the evidence does not observe, or in material already released. We investigate three questions. **RQ1:** How should changes in representation, roles, relationships, and participation be expressed as decisions about management, receipt, restriction, and retention? **RQ2:** What guarantees and service costs do authorization caching, revision checks, and scoped encryption provide for those decisions? **RQ3:** What evidence supports a stated transition claim, and which judgments or residual exposures remain open?

Our contributions are:

- A CI-grounded transition model and decision table that separate management, receipt, restriction, and purpose-specific retention, tied to UK data-protection conditions (Sections 2 and 4, Table 1).
- A timeline analysis of exposure: an analytical staleness bound for leased authorization, a budget-indexed comparison of 46 configurations, matched policy histories, and a constructed hidden-path replay. Together they show that once cache staleness is bounded, exposure remains in decision intervals, unobserved paths, and released material (Section 6, Table 7).
- Claim-scoped EvidenceRecords, generated from saved traces, that link a classified requirement to declared queries, paths, and times, and report coverage gaps and unfinished actions alongside errors (Sections 5.2 and 6.5).

Co-design and social goal. Legal classification determines which operations a request should stop, for whom, and on what clock; the experiments show that such a clock is assessable only with a query scope, path inventory, and interval. We design toward young members' evolving control over their contact information and members' ability to end unwanted promotional contact, while preserving justified record-keeping and independent services.

Reading guide. The legal argument is in Section 2, Table 1, and Section 7; the systems argument is in Sections 4–6, especially Tables 2 and 7.

Scope and evidence. We stipulate processing in the context of a UK establishment and use England and Wales for children's exercise of rights. Public documents establish stated institutional rules; legal examples supply conditional requirements; proposed defaults are policy choices. The experiments are finite synthetic executions, without platform access, real member records, or an assessment of the Church's actual compliance. An episode is a workload instance, not a person or an independent statistical sample. Section 8 states the trust and generalization limits; later sections do not repeat them.

2 Legal and Institutional Context

2.1 Territorial Scope, Controllers, and Internal Authority

The UK GDPR combines religious-data conditions, individual rights, design duties, and accountability. We stipulate processing in the context of a UK establishment under Article 3(1), which does not depend on processing location [3, art. 3]. This is a jurisdiction-specific design analysis, not a determination of the institution's territorial coverage. Other jurisdictions require fresh analysis of grounds, deadlines, and remedies; the system distinctions between management and receipt, retention and use, and server updates and recipient material carry over.

Controllership turns on determining processing purposes and means, including joint determination, rather than an administrator's ability to edit a field [3, arts. 4(7), 26]. The Church's Privacy

Notice identifies its Utah corporation sole and other Church entities as responsible joint controllers, and ordinarily describes local wards and stakes as subdivisions rather than separate legal entities [2, sec. 1]. In *Jehovan todistajat*, C-25/17, the CJEU considered a religious community's organizing and coordinating role in members' collection of personal data; access to all collected data was not necessary to joint control [4, paras. 68–75]. That judgment under Directive 95/46 supports a functional inquiry into the particular processing arrangement, not a conclusion that every volunteer is a controller.

We therefore identify a synthetic responsible controller U and separately model the authority to approve a particular change. A household representative, S, or an authorized case handler can supply an approved decision within a mandate without replacing U's accountability. This keeps a technical *manager* field from silently answering a legal question about responsibility.

2.2 Religious Data, Recipients, and Continuing Purposes

Contact information linked to membership can reveal religious beliefs. Processing therefore needs both an Article 6 lawful basis and an applicable Article 9 condition. Article 9(2)(d) permits specified processing in a qualifying religious not-for-profit body's legitimate activities, with appropriate safeguards, concerning members, former members, or regular contacts connected with its purposes. Reliance on that condition also requires that personal data not be disclosed outside the body without the data subject's consent [3, arts. 6, 9]. It supplies neither a general Article 6 basis nor an unlimited internal audience.

Our legal examples examine optional directory visibility under Article 6(1)(f), subject to necessity and balancing, and promotional email under Article 6(1)(a) consent. These are analytical premises, not the institution's established bases; Section 7.1 shows how the basis changes request classification. Direct marketing is a purpose that can overlap with consent or another basis. Article 6(1)(ea) has separate statutory conditions and is not interchangeable with ordinary legitimate interests. The prototype executes supplied policies; it does not select their legal grounds.

The recipient's capacity, activity, and safeguards matter: a private telephone does not establish disclosure outside the body, and membership does not justify every use. The Handbook distinguishes resignation from a no-contact request [5, sec. 32.14.9]; the Privacy Notice describes continuing records and rights routes [2, secs. 7–8]. Article 9(2)(d)'s inclusion of former members permits no inference about every purpose, duration, or recipient: purpose and storage limitation, conditional erasure, and restriction still apply [3, arts. 5, 17–18]. Institutional guidance addresses printed and electronic record protection and prohibits exporting membership or financial data from Church applications [6, 7]; our system-supported copy/print fixtures test release scope and residual material, not a deployed export feature.

2.3 Children's Rights, Representation, and Service Scope

Data protection rights belong to children. In England and Wales, understanding matters to exercising those rights, and representation should be reviewed as the child develops [8]. S's ability to explain the requested audiences and consequences is an explicit assumption, not a measured assessment; it supports review at 17, and adulthood is a separate administrative trigger. Article 8(1)'s UK age of 13 concerns consent-based information society services offered directly to children, not general directory-control transfer [3, art. 8].

Article 25(1) requires appropriate measures at design and during processing. Bygrave emphasizes its organizational as well as technical reach and the difficulty of specifying its engineering requirements [33, pp. 114–118]. Article 25(2) addresses purpose-specific necessity, accessibility, and storage. Whether a ward audience is an "indefinite number" of persons is a factual question [3, art. 25].

Articles 25(1A)–(1B), in force from 5 February 2026, add children's higher protection matters for information society services likely to be accessed by children. Article 25(5) excludes preventive or counselling services, and paragraph (4) states that paragraphs (1A)–(1B) imply nothing about appropriate measures in other cases, so the general paragraph (1) duty remains relevant [3, art. 25]. A directory's ISS status depends on the ordinarily economic service concept, not on "app" or nonprofit status. The Children's Code also needs a service-scope assessment; DPA 2018 sections 123 and 127 give it statutory and evidential significance, without liability for nonobservance alone [9, 10]. We do not determine either regime's application to the actual directory.

The Code's statutory framework takes account of UK UNCRC obligations. General Comment 25 joins evolving capacities, parental guidance, privacy, and agency [11, paras. 19–21, 85–86]; supporting S's choice need not imply that all parental involvement is inappropriate. Livingstone et al. document developmental and contextual variation in children's privacy understanding, which supports review of a specific choice rather than a birthday-based competence inference [35, secs. 5.3, 5.9–5.10].

Responsibility for reviewing representation. We propose a designated handler with an explicit mandate, preservation of the child's request, accessible explanation of audiences and consequences, recorded reasons, and further human review of a refusal or disputed mandate. This is a governance proposal, not a universal capacity test. Fixtures supply understanding and handler authority; a deployment must justify both, which age or a boolean flag cannot do. Parental disagreement and safeguarding require separate procedures (Section 8).

2.4 Accountability and the Limits of Technical Evidence

Articles 5(2), 24, and 32 ground questions about demonstrating compliance, implementing appropriate measures, and risk-sensitive security [3]. They prescribe no record format. A denial log can support a claim about that response; a claim about all affected services needs a justified inventory and wider evidence. Appendix A.9 notes the human-rights context.

3 Related Work

Contextual integrity. Nissenbaum frames privacy as the appropriateness of flows relative to contextual norms [14]. Barth et al. formalize such norms in a temporal logic over flows among agents acting in roles [15]. We adopt CI's flow parameters and ask a narrower operational question: when the context changes, which norms change, by when must the change take effect, and what evidence shows that it did?

Formalizing privacy requirements. Nissim et al. examine translation between legal and computational privacy standards; Cohen and Nissim formalize singling out [16, 17]. We likewise expose interpretive assumptions, keeping classification, capacity, and legal sufficiency as review inputs while formalizing conditional execution claims.

Design duties and effective rights. Veale, Binns, and Ausloos show how confidentiality-oriented design can obstruct access, erasure, and objection, and argue for explicit tradeoffs and systems that support rights [34]. Our case examines a complementary tension: a retained record, a valid service grant, and a fresh authorization check can coexist with an unresolved objection or an unjustified audience. Separating these decisions makes their implementation assessable without treating encryption as a complete rights remedy.

Dynamic authorization and encryption. Relationship-based models address cascading revocation and provenance [18]. Zanzibar's *new enemy problem* includes old access rules applied to new content; zookies link freshness and content ordering [19]. Our changed contact values instantiate that

established concern. Plutus separates later-version protection from already available material [20]; key-rotation research studies cryptographic updates [21]. We propose no new rotation construction or cooperative recipient deletion.

Deletion, databases, and the institutional setting. Deletion definitions make system and interaction assumptions explicit [22]; GDPR database benchmarks examine implementation costs [23]. We ask which uses a justified retained record still permits and what transition evidence exists. Studies of adolescent portals and religious tracking motivate the setting [24, 25] but supply no observations of this directory's transitions.

4 Model and Protection Claims

4.1 State and Reasoned Transitions

Let a policy revision be $\Sigma^k = (F^k, D^k, G^k, B^k, H^k)$. Facts F include age, active relationships, account status, and their supplied provenance. D specifies who can manage particular settings within a mandate. Recipient grants G specify actor, subject, fields, operations, purpose, validity interval, and continuing dependencies. Restrictions B override affected grants. Retention scopes H record field, purpose, reason, allowed operations, status, and review point. Retention supplies no recipient grant.

For request $q = (u, s, f, o, p)$ and modeled path m, ordinary recipient permission is

$$P_\Sigma(q, m, t) = \mathsf{Grant}(G, F, q, m, t) \wedge \neg\mathsf{Block}(B, q, m, t) \wedge \mathsf{Compatible}(H, q, m, t). \tag{1}$$

In CI terms, u, s, and f are the recipient, subject, and information type; the directory service on path m is the sender; and the operation o, purpose p, and the conditions of the matching grant act as the transmission principle. A transition changes Σ^k, and thereby the norms that Equation 1 enforces. Grant requires a matching live grant and its continuing dependencies; hard validity is $t_{\text{start}} \le t < t_{\text{expiry}}$. Compatible is true in the ordinary service phase and otherwise requires the field, purpose, and operation to fit an applicable retention scope. Management checks D separately, and the identity-bound rights route is separate from directory privileges. Reading implies neither management nor a system-supported copy or print operation, and withholding those operations cannot prevent manual copying of displayed plaintext.

A transition object is $x = (e, \pi, d, A, \mathcal{T})$: event, supplied facts and provenance, reasoned decision, authorized actions, and time record. Define the partial update

$$\delta(\Sigma^k, x) = \mathsf{Apply}(\Sigma^k, A) = \Sigma^{k+1} \quad \text{provided } \mathsf{Authorized}(d, D^k, F^k) \wedge \mathsf{Consistent}(A, d). \tag{2}$$

The authorization premise concerns the decision-maker's mandate, not the decision's legal sufficiency. The prototype trusts supplied decisions, validates supported actions, rejects duplicate events, and rejects superseded snapshots. Table 1 states the decisions the model must distinguish, answering RQ1.

4.2 Time, Scope, and Evidence

The time record distinguishes event t_e, receipt t_r, classification/review t_l, each commit t_c, and requirement deadline t_d. Interim protection can precede classification, and a late commit does not reset an overdue requirement. Figure 1 shows S's history. Legal time rules are set independently of implementation timing.

Give an assessable requirement a stable identifier ρ, request set Q, path inventory M, effective time, deadline t_d, and end of applicability t_{end}. Let $\Sigma_\rho(t)$ denote the operative target policy for that

Table 1. Transition decisions. Outcomes are conditional on facts and reasons; proposed defaults are identified. Several state components may change together.

Trigger	Decision and management	Receipt, retention, and review	Legal anchor / open input
Under-18 self-request	Review understanding and representative mandate; update D if justified	Review inherited G and B separately; optional sharing confirmation is our proposal	Child's rights; Arts. 12, 25; capacity is supplied
Adult review	Revisit D and reasons, rather than create rights at 18	Preserve, narrow, or reconfirm each audience with reasons; no automatic new G	General design duties; actual service and facts
Role or relationship ends	Record who can commit the change	End dependent G; preserve an independently justified grant; reassess object scope	Arts. 5, 6, 9, 25, 32; dependency and purpose
No-contact request	Preserve raw request; classify purpose, channel, basis, and rights; interim action may precede classification	Marketing objection stops that purpose; qualifying restriction limits operations, including on retained data	Arts. 7, 18, 21; PECR 22; see Section 7.1
Departure or closure	Keep a route to U after directory closure	Review ordinary G, B, and H separately; retention requires purpose, scope, and review	Arts. 5, 9(2)(d), 12, 17–19; no universal erasure inference
Superseding decision	Validate order, mandate, and decision identifier	New $G/B/H$ replace only their stated scope; late older state must not restore access	Arts. 24, 32; ordering and coverage evidence

requirement at time t. A release claim is

$$\begin{aligned}\mathsf{Claim}(\rho, Q, M, [t_d, t_{\mathrm{end}})) : \quad & \forall q \in Q\ \forall m \in M\ \forall t \in [t_d, t_{\mathrm{end}}), \\ & \mathsf{Serve}_m(q,t) = 1 \implies P_{\Sigma_\rho(t)}(q,m,t) = 1.\end{aligned} \tag{3}$$

A classification or unrelated grant can create a revision without ending the requirement or restarting its deadline; a scope-changing decision must explicitly supersede or subdivide it. Observations bounded by one revision therefore need not cover the requirement's full assessment window. Claims also concern recoverable fields and versions: an email entitlement cannot justify a bundled restricted telephone number. Permitted-operation metrics are reported separately, since universal denial satisfies Equation 3 while preventing service.

Decision evidence explains authority and scope; execution evidence identifies outcomes, paths, versions, and times; residual evidence accounts for released material. Their extent follows the claim: a single refusal needs less evidence than a broad transition assessment.

4.3 A Bound on Authorization Staleness

Consider an integer-tick sequential gate, an old decision cached at r, a permission flip committed at t_c, and a lease of length $\tau > 0$. Assume a fresh evaluation uses current truth, the permission remains flipped, and an expired entry is not extended during an outage. Superseded responses can

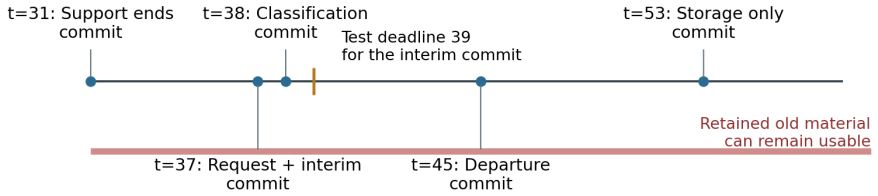


Fig. 1. Part of the executed synthetic history running_S. The request and interim block occur at 37; classification follows at 38. The marked deadline is the experiment's two-tick budget for the interim commit, not a legal grace period. Previously released material can remain usable across subsequent transitions.

then occur only at

$$t_c \le t < r + \tau \text{ (strict lease)}; \qquad t_c \le t \le r + \tau \text{ (inclusive lease)}. \tag{4}$$

The bound follows by intersecting the entry's reuse interval with the period after the flip; known grant boundaries or pushed invalidation can shorten it. It covers superseded allows and denials, not fail-closed outage denials.

Define a late false grant by $t \ge t_c + b$. In the main experiment events precede all probes at a tick, so an old fill has $r \le t_c - 1$. Sufficient budgets are then $b \ge \max(0, \tau - 1)$ for strict leases and $b \ge \tau$ for inclusive leases. If a same-timestamp fill can precede the commit ($r = t_c$), they become $b \ge \tau$ and $b \ge \tau + 1$. Setting $\tau = 0$ always reevaluates. Under the main ordering, strict $\tau = b + 1$ lies at the edge of the sufficient region for integer $b \ge 0$. Strict $\tau = 1$ meets $b = 0$ by reusing a decision only within its fill tick, which in Table 2 saves just 10.5% of full evaluations. A zero budget thus leaves little room for lease-based caching, though it does not by itself require the guard.

4.4 Released Material and Encryption

Observation. If a recipient retains a plaintext value, or a usable decryption key with its ciphertext, and no trusted deletion mechanism governs that state, then no server-side change of authorization, keys, or current ciphertext removes the recipient's access to the old value: the recipient keeps the plaintext, or reruns decryption locally on unchanged inputs. This is the familiar boundary behind revocation and deletion work [20, 22]. In CI terms, encryption can bind later values to the new context's norms, but it cannot undo a flow that the earlier context permitted. It does protect a new value encrypted under an independent key the former recipient never received, which is what makes key and release scope a design decision rather than an afterthought.

5 Mechanisms and Evidence Implementation

5.1 Authorization and Scoped Encryption

The single-process prototype accepts approved transitions and assumes bound identities. A TTL gate caches positive and negative decisions; $\tau = 0$ reevaluates. A revision guard reuses an entry only after checking the authoritative revision and known temporal boundaries; otherwise it reevaluates. An unavailable required check withholds the operation. Push messages evict older entries after three ticks, retrying during lookup unavailability. A push-only control omits the guard.

The field-store configuration, M_2, adds AES-256-GCM with independent field keys, 12-byte nonces, and 16-byte authentication tags [26, 27]. It refreshes a field's key when the content or its eligible readership changes. A surviving read grant through another purpose remains part of the readership, while a change of management alone need not rekey. Associated data binds an

opaque object identifier, field scope, content version, and key revision. Key release checks current authorization and the scope of every field in the object. U is trusted with plaintext and keys.

Purpose and operation labels constrain supported workflows, not memory or later human use, and an independent valid email entitlement can preserve knowledge after another purpose ends. Encryption and release scopes must therefore align: a coarse bundle can disclose a restricted field or block a legitimate narrower task.

5.2 Enforceable Retention and Generated Records

The original eight histories used retained-field metadata. The extension adds typed entries for field, purpose, reason, allowed operations, status, review tick, and deletion status. After departure, an administrative email grant must also pass H in Equation 1; a later storage-only scope prevents release despite that grant. Physical erasure is not implemented. Management and rights operations are separately controlled.

The EvidenceRecord generator reads saved fixtures, commit observations, request traces, and linked residual observations, and performs a deterministic transformation rather than calling the policy evaluator. Records preserve the raw request and supplied classification; responsible controller and decision actor; event, receipt, classification, commit, and test-deadline times; target and observed decision revisions; affected and observed query sets; a half-open observation interval; response reasons; the last observed stale allow or `null`; and unfinished actions. A zero late-error count is always reported with the number of post-budget probes, so zero coverage cannot pass as successful verification. Acknowledgment is unmodeled, possible notification is recorded as unexecuted, and recipient deletion and backups as unverified. The member view offers a shorter explanation and review route; it has not been evaluated with users.

A separate scope validator receives declared query, path, time, and conditional action requirements independently of the observed probes, and a requirement-level sidecar links revisions without resetting the deadline. Section 6.5 reports both.

6 Evaluation

6.1 Workloads, Metrics, and Checks

The main scan crosses eight histories, five seeded query orders, two lookup conditions, and 46 configurations: 3,680 episodes and 2,701,120 decisions. Each episode lasts 60 ticks and issues $26 + 59 \times 12 = 734$ queries: all 26 types initially, then six fixed and six sampled probes per tick. Queries cover reading, management, supported copying and printing, and an identity-bound rights route. The histories cover two representation defaults, appointment end, relationship transfer, two no-contact timings, departure/retention, and superseding changes; Appendix Table 11 reports every history.

The fixed manifest includes 21 TTL values, two endpoints, and nine budgets: 42 TTL configurations, two guards, and two push-only controls; budget reclassification creates no new executions. Lookup is uninterrupted or unavailable at ticks 14–16 and 36–38; content and the rights route remain available. Events precede probes, and hard grant validity is half-open. Changed values let stale decisions reveal subsequent versions. The reporting budget $b = 2$ starts at each permission flip, not at request receipt.

False grants (FG) allow a query denied by its supplied reference policy; false denials (FD) reject a reference-permitted query. We separate authority-unavailable denials (FD_U, a deliberate fail-closed choice) from superseded negative-cache denials (FD_C). Late FG is measured at or after the permission-change tick plus b. *Promotional-outreach probes* are query `N_EMAIL_OUTREACH`: field

Table 2. Budget-indexed choices under the main event ordering. Strict $\tau = b + 1$ is the edge of the sufficient region, not an optimum. Counts combine both lookup conditions. FD_U: authority-unavailable withholding; FD_C: superseded negative-cache denial. Reduction counts full evaluations only.

b	Gate	FG	Late FG	FD_U/FD_C	Full eval.	Reduction
–	Fresh	0	0	1,172/0	49,264	0.0%
0	$\tau = 1$	0	0	1,172/0	44,080	10.5%
1	$\tau = 2$	106	0	1,021/36	29,016	41.1%
2	$\tau = 3$	254	0	721/102	22,544	54.2%
5	$\tau = 6$	612	0	569/194	14,312	70.9%
10	$\tau = 11$	712	0	142/182	9,400	80.9%
all	Guard	0	0	1,172/0	6,056	87.7%

The guard additionally performs 51,744 revision checks, a different work unit.

access for the stipulated fundraising/promotional email workflow; no email is sent. Rights requests are counted separately.

Separately written relation and cache interpreters matched every scan decision and work counter and detected all specified wrong implementations; boundary cases check exact error-tick sets, and the retention schema has its own randomized comparison (Appendix A.4). The final isolated Python 3.12.14 environment (`cryptography` 50.0.2) passed 80 tests and reran the boundary cases, new histories, evidence generation, cryptographic controls, and 80 M_2 baseline episodes; the earlier scan is retained with matching source hashes and trace reaggregation rather than counted as rerun.

AI-assisted research workflow. OpenAI ChatGPT/Codex assisted research scoping, synthetic fixtures and tests, Python mechanisms and result processing, plotting code, and argument revision. Anthropic Claude assisted with research-design feedback (including the budget-indexed comparison, the request-classification table, and the hidden-path replay design), checks of legal sources, and execution of the hidden-path replay. No real member data was generated. The checks above address implementation errors within this shared AI-assisted workflow, not external validation, and legal propositions were checked against source texts without independent legal review. The author is responsible for verifying code, sources, interpretations, and outputs.

6.2 Freshness, Availability, and Query Work

Budget-indexed comparison. Table 2 pairs each budget with the largest strict lease in the sufficient region. At $b = 0$, strict $\tau = 1$ reduces full evaluations by 10.5%; at $b = 2$, strict $\tau = 3$ has 254 within-window FG, zero late FG, and 102 stale-cache denials, while saving 54.2%. The guard performs 6,056 full evaluations plus 51,744 revision checks; comparing the two units requires deployment cost assumptions.

The cache's service effects run in both directions. TTL 30 permits 3,068 disallowed operations, 2,738 of them after the budget, and denies 1,040 allowed ones; 1,020 of those denials come from superseded negative entries and only 20 from unavailable authority. Guard denials have the opposite explanation: all arise from its availability policy. Strict push-only still has 172 late FG, so push does not substitute for enforcement in this workload; guard with and without push have identical outcomes and work. Appendix A.5 separates lookup conditions and FG types; TTL 30's largest sampled FG offset is 22 ticks.

A finite workload can also show zero late FG outside the sufficient region when sampled queries miss an error window (Appendix Figure 3); such cells do not strengthen Equation 4. The 26,072

Table 3. New-schema histories: FG / late FG / FD_U / FD_C. Each cell covers 7,340 decisions; late FG uses $b = 2$. The under-18 subject remains 17 throughout that history.

History	Fresh	TTL 3	Guard + push	Push only
Under-18 request	0/0/210/0	6/0/121/0	0/0/210/0	4/0/9/22
Running case S	0/0/199/0	51/0/118/17	0/0/199/0	249/81/26/64

Table 4. Retained-material combinations, each checked in all 1,000 randomized instances with distinct old and new values and independent keys.

Material supplied	Old ciphertext	New ciphertext
Old key	Old value recovered	Authentication rejected
New key	Authentication rejected	New value recovered

boundary cases check both permission directions, endpoints, fill/commit orderings, and critical budgets.

Per history (Appendix Table 11), the guard and Fresh have no FG, and TTL 3 has none after the budget; delayed superseded snapshots are rejected. The M_2 replay reproduces the guard's online outcomes by construction, so it adds cost data rather than independent evidence of online safety: 750 encryptions, 20,954 key releases/decryptions, 23,080 key-service policy queries, and 32,440 readership-maintenance queries, none of which appear in the gate columns.

6.3 Under-18 Choice, Retention, and the Running Case

The new-schema workload comprises two histories, four configurations, five seeds, and both lookup conditions: 80 episodes and 58,720 decisions. In the under-18 history S remains 17, requests review at 7, receives settings control and narrower optional sharing at 9, and confirms sharing at 23; existing support and event grants survive the management change. The combined running history adds adult review at 19, relationship reassignment at 31, interim contact restriction at 37, classification at 38, departure at 45, an independent rights request at 47, and storage-only retention at 53. It is a single executed history.

Table 3 shows that strict TTL 3 again has no late FG, while the push-only running case has 81. At tick 53 the retained administrative email cannot be read even though its underlying grant survives: H restricts operations rather than merely labeling a field. The rights-request route remains available after account closure; completion of a rights response and physical deletion are outside the model.

Matched handover interval. The two original authority histories, with guard-with-push and uninterrupted lookup over $[7, 19)$, pooled across five seeds, permit 12 of 12 optional directory-phone probes under inherited sharing versus 0 of 12 under narrowing. Both permit all 167 sampled independent service operations. The 12 withholdings under narrowing follow that policy and are not FD; each history has zero FG/FD in this interval.

6.4 Retained Keys, Versions, and Release Scope

Table 4 executes all four combinations. Each instance also rejects an email key against a telephone object and rejects modified associated data or ciphertext. A positive control discloses a new value when it is intentionally encrypted under the recipient's known old key with a fresh nonce—key reuse across an authorization boundary, not GCM nonce reuse. A coarse bundle discloses the restricted telephone to an email-only recipient, while strict bundle-scope enforcement also blocks

Table 5. Condensed fields from a generated record. “Unknown” and “unobserved” delimit the claim.

Field	Recorded value
History / event	`running_S` / support relationship ended
Revision / observation interval	5 / [31, 37); test deadline 33
Changed / observed labels	4 / 3; `N_ADDRESS_SUPPORT` unobserved
Responses / post-budget	14 / 9; FG = FD = late FG = 0
Last observed stale allow	`null`; all-services-applied revision unknown
Residual / remaining actions	5 local material observations; possible notification unexecuted; recipient deletion unverified

Table 6. Evidence checks. Upper rows: saved traces from the main scan with a declared bounded control and four scope/action challenges. Lower rows: a constructed hidden-path replay (Appendix A.8). Path and notification requirements are declared inputs.

Case	FG/FD/late FG	Separate finding
Declared bounded control	0/0/0	Declared checks satisfied
Additional required query unobserved	0/0/0	Query coverage gap
Additional required path unobserved	0/0/0	Path coverage gap
Window ends before deadline	0/0/0	No post-budget coverage
Required notice stipulated, unexecuted	0/0/0	Required action unfinished
Inventory scope, primary path probed	0/0/0	Secondary path unobserved
Secondary path also probed	5/0/5	Stale grants on all 5 secondary probes
Secondary path repaired (guard)	0/0/0	Declared checks satisfied

the valid email task. All controls produce their expected result in every instance (5,000 encryptions, distinct key/nonce pairs). These are implementation checks of the observation in Section 4.4, not an AEAD security proof.

Separating three fields costs 84 bytes of nonce/tag overhead and 96 bytes of keys, against 28 and 32 for one object. In S’s local field-store episode, 105 event/material observations show old pairs remaining decryptable after the relevant readership ends, while retained old keys fail on current packets.

6.5 What an EvidenceRecord Establishes

The 13 generated records cover receipt, classification, review, and authorization changes. Table 5 condenses S’s relationship-end record: three of four affected query labels receive 14 probes, nine after the test deadline, without FG or FD, and five local material observations are linked. The record’s value lies in what it marks as open: an unobserved label, an unknown all-services revision, and unexecuted notification.

The interim promotional-contact record exposes a different coverage issue: its revision window is [37, 38) and ends before the test deadline 39, so its zero late FG comes with zero post-budget probes. Classification at 38 does not end the contact restriction; assessment must follow the restriction across later revisions.

Continuing requirement. The sidecar retains test deadline 39 across revisions 6–10 over [37, 60), linking 5 original records and 28 responses, 26 of them post-budget, without FG or FD. Classification does not restart the deadline.

Table 7. Where exposure arises along a transition, and what governs it in this model.

Location	Governed by	Observation	Source
Before classification, $[t_r, t_c)$	Request classification and interim action	Review-first permits 73/73 promotional-outreach probes, interim restriction 0/73; inherited sharing permits 12/12 optional directory-phone probes, narrowing 0/12	§6.3, §7.1
Enforcement window, $[t_c, t_c + b)$	Lease length within budget b	Strict $\tau = 3$ at $b = 2$: 254 FG inside the window	Table 2
After the budget, $t \geq t_c + b$	Mechanism correctness	Strict $\tau = 3$ and guard: 0 late FG; push-only: 172; strict TTL 30: 2,738	Table 2, §6.2
Unprobed path	Declared evidence scope	Secondary path: late FG on 5/5 probes while primary-path evidence reports 0/0/0	Table 6
Released material	Nothing after release	Old key with old ciphertext recovers the old value	Table 4, §4.4

Evidence challenges. In the upper rows of Table 6, every selected response set has FG = FD = late FG = 0, yet separate checks expose an unobserved required query, an unobserved required path, zero post-budget coverage, and a stipulated notice left unexecuted. A gap of this kind marks the claim as unestablished; it does not by itself show a fault. Conversely, "declared checks satisfied" means only that the declared sampled checks passed: observing every query label and path at least once does not cover every query–path–time combination in Equation 3, which would require exhaustive observation, an analytical bound such as Equation 4, or further assumptions.

Hidden-path replay. The lower rows show that such a gap can hide a real fault. A separate constructed replay adds a secondary path that is listed in the deployment inventory but omitted from the initial probes, and that lacks revision checks and push invalidation (strict $\tau = 30$, filled one tick before the commit at 7; $b = 2$; window $[7, 14)$). Evidence scoped to the primary path reports 7 responses with no FG or FD; the inventory-scoped contract flags the secondary path as unobserved. Probing it from the deadline yields late false grants on all 5 secondary probes, as Equation 4 predicts for $\tau > b + 1$, while adding those probes leaves primary-path responses unchanged and a guarded secondary path yields none. The fault is constructed; the point is that zero observed errors on the probed path carried no information about the unprobed one. The validator detects the gap only because the inventory was declared independently.

6.6 Where Exposure Arises

Table 7 places each result on the transition timeline. The counts come from different workloads, so the table compares where exposure arises, not its magnitude.

Under the evaluated ordering, a correctly sized lease or a revision guard closes the post-budget row, and the budget bounds the enforcement window. The remaining rows are closed, if at all, by classification rules, a declared evidence scope, and limits on what is released in the first place. These are the points at which law and audit, rather than mechanism choice, determine the outcome. In CI terms, the mechanism enforces whichever norm it is given; whether the norm changed in time, everywhere it should, and before the information left, is decided elsewhere.

Table 8. Conditional request classification. The branches specify legal premises for policy inputs; the software does not decide that a basis or exception is established.

Premise for the processing	Consequence to assess and record
Consent, Art. 6(1)(a)	Withdrawal under Art. 7(3) ends reliance on that consent. Art. 17(1)(b) concerns erasure where no other legal ground exists, subject to Art. 17(3). Earlier lawful processing is not retrospectively invalidated.
Ordinary legitimate interests, Art. 6(1)(f)	An Art. 21(1) objection concerns the person's particular situation and the controller's possible compelling overriding grounds or legal claims. Art. 18(1)(d) gives a right to restriction pending that verification.
Art. 6(1)(e) or (ea)	Art. 21(1) also names these bases, but their own public-task or recognized-interest conditions must first be established; neither is assumed for this directory.
Direct-marketing purpose, across bases	Under Art. 21(2)–(3), the objected-to processing must no longer occur for that purpose. PECR imposes a separate email-channel rule.
Unclear purpose, basis, or request	Preserve the request and uncertainty, seek appropriate clarification, and record the grounds and scope of provisional protection. "Under review" is not a new lawful basis.

7 Legal and Policy Implications

7.1 No-Contact Requires Classification Before Mechanism Comparison

A request need not cite a provision [28, 29]. Preserve its words, sender or representative basis, receipt time, processing basis, purpose, channel, handler, uncertainty, and interim action. Table 8 separates basis from purpose: withdrawal and a marketing objection can coexist.

An obtained Article 18 restriction limits processing beyond storage to paragraph (2)'s cases, and paragraph (3) requires informing the person before lifting it. An unclassified request does not automatically prohibit every purpose, and consent withdrawal does not justify silently substituting an unestablished alternative basis [3, arts. 6–7, 17–18, 21].

PECR regulation 22 separately governs the stipulated unsolicited fundraising/promotional email to an individual subscriber. Regulation 22(3A)'s charitable soft opt-in, in force from 5 February 2026, requires statutory charity status, collection when the person expresses interest in or supports the charitable purposes, the sole purpose of furthering those purposes, and a simple free refusal at collection and in each message. ICO guidance limits it to details obtained on or after that date, including qualifying fresh provision by an existing supporter. We establish neither status nor conditions for the stipulated entity. Prior consent is needed unless an exception under regulation 22(3) or (3A) is satisfied, and neither overrides a refusal [30, 32].

Requirements accumulate: an Article 6 basis does not satisfy PECR, a PECR exception does not supply an Article 9 condition, and an outside disclosure relying on Article 9(2)(d) also needs its consent condition. Routine pastoral care is not marketing merely because a religious body initiates it.

The matched no-contact histories isolate decision delay. With guard-with-push and uninterrupted lookup, pooled across five seeds, review-first permits 73 of 73 promotional-outreach probes during $7 \leq t < 19$, versus 0 of 73 under interim restriction. Both follow their own policies exactly, so execution evidence alone cannot distinguish them. Under a clear marketing-objection premise, correct later enforcement cannot justify the earlier continuation; deciding whether a policy satisfies

the request requires a requirement ρ declared independently of the policy being executed, which is what an EvidenceRecord checks against. The counts are field operations, not established violations.

Article 12A's one-month response period, running from its defined relevant time [3, 31], is no marketing grace period, and Article 17 separately requires erasure without undue delay where it applies. A record should preserve receipt, interim action, classification, commit, and the independently justified deadline.

7.2 Youth Representation and Continuing Use After Departure

S's request separates exercise of rights, representation, settings control, inherited sharing, and independent service grants. Review need not wait for a birthday, but the handler must justify understanding and authority rather than infer them from account status.

We propose an accessible choice about optional broad sharing at handover, with narrowing pending confirmation as one option. Refusal should be unpressured, unrelated services independent of wider sharing, and choices revisable. These are design proposals; the handover counts show their access consequences, not which outcome a family dispute should reach.

Departure separates account access, contact, retention, and continuing rights. Former-member status under Article 9(2)(d) does not authorize new outreach; each use needs a purpose, lawful basis, special-category condition, necessary scope, and safeguards. A minimal suppression record can help honor a refusal but needs its own restricted justification. Reducing disclosure and enabling rights therefore require coordinated design [34]. Our administrative retention reason is a reviewable synthetic assumption.

Article 19 notification applies subject to its impossibility and disproportionate-effort exception, and Article 18(3) requires informing the person before lifting a restriction. Notice, acknowledgment, and removal are separate actions. The independent rights route tests request availability after closure, not a completed response or identity review.

7.3 From Legal Duty to an Assessable Claim

Table 9 connects RQ2 and RQ3. A narrow denial claim can rest on a response record. Assessing that a former support worker's relevant access ended also requires the operative decision and dependencies, the relevant paths and versions, and the material already released.

An EvidenceRecord can expose missing observations or unfinished actions despite zero observed errors. An unexecuted notice is a required unfinished action only if the duty's applicability is established. Scope and duty inputs must be declared independently of the observed probes; otherwise omission defines itself away, as the hidden-path replay shows.

Auditability also consumes privacy: record access, retention, and identifiers need role- and purpose-specific limits.

8 Limitations, Validity, and Research Practice

Trust and validity. The controller, decision service, identities, mandates, event ordering, and experimental logs are trusted. Former recipients retain material but do not compromise the service or collude for fresh keys. The model excludes dishonest logs, complete mobile and backup discovery, distributed check-to-release atomicity, clock skew, and end-to-end key delivery. Logical ticks, including the budget b, have no legal-duration interpretation.

Workload counts follow from designed probes, and successful decryption does not measure social harm. One institution's public documents and synthetic histories support no prevalence or deployed-performance estimate. Children's understanding, lawful grounds, retention reasons, and notification duties are review inputs. No independent UK legal reader, youth consultation, or institutional stakeholder review has been obtained. Parental disagreement can challenge who may

Table 9. Legal questions, proposed system properties, evidence, and residual issues. A conditional evidence design, not a statutory checklist.

Legal question	Technical property / state	Evidence artifact	Residual issue
Child's rights; Arts. 12, 25	Review F, D, G, B; separate management and receipt	Request; understanding, mandate, reasons, changed scopes	Capacity, interests, disputed mandate
Arts. 6, 9(2)(d): purpose and recipients	Purpose/operation scopes in G, B, H; field-scoped release	Basis, condition, recipient capacity, fields and versions	Body boundary, later misuse, copies
Arts. 7, 18, 21; PECR 22	Classify rights; scoped restriction and time rule	Request, basis, purpose/channel, interim action, commit, coverage, notices	Unclear requests, exceptions, other channels
Arts. 5, 17–19: retained records	H creates no G; storage-only scope; rights route	Purpose/field reasons, allowed operations, review and action status	Retention necessity, erasure, recipient action
Arts. 24, 25, 32: implementation	Revision checks, bounded caches, scoped keys	Target/used revisions, FG/FD reasons, declared path inventory, hashes, material witnesses	Log authenticity, races, collusion, unknown copies

decide, and safeguarding can justify narrowly different information flows; those cases need distinct reasons, restricted disclosure, and review procedures rather than a universal birthday rule or an assumption that either parental access or its removal always prevails.

Ethics and provenance. The study uses public institutional documents and synthetic data. It involves no recruitment, interviews, observation of individuals, real member records, private communications, account logins, or interaction with institutional systems. The author's familiarity with the setting motivated the scenarios but supplies no empirical sample. The legal-source cutoff is 18 October 2026 (UTC).

Reproducibility. The supplementary package contains the new-schema data, evidence checks, saved summaries, and hidden-path replay; the main-scan and cryptographic scripts are not included (Appendix A).

9 Conclusion

When a context changes, access must change with it: a youth request can change management without canceling independent service grants, a contact objection can require action before review ends, and retention can permit storage without use. Within the stated model, a correctly sized lease or a revision guard confines mechanism staleness to a budget, but exposure remains in the classification interval, in paths the evidence does not observe, and in material already released. Encryption binds later values to new norms but cannot recall an earlier flow. Claim-scoped evidence records make each location inspectable and keep visible the legal judgments and unobserved paths that technical success cannot settle.

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A Reproduction and Coverage

A.1 Files and Commands

The supplementary package for this submission contains: the fixtures, commit observations, request traces, residual observations, and generated EvidenceRecords for the two new-schema histories (`artifact/`); the evidence validator, its eight controls, the scope/action challenges, and the continuing-requirement sidecar, with their logs and outputs (`revision_06/`, `reports/evidence_checks_run/`); the saved budget and handover summaries and the scatterplot data (`reports/saved_postprocessing/`); and the hidden-path replay with its outputs (`hidden_path/`). The evidence checks and the hidden-path replay can be rerun with the standard library:

```
python3 run_remaining_checks.py
cd hidden_path && python3 -m unittest -v test_hidden_path.py
python3 run_hidden_path.py --output results_rerun
```

The scripts for the 46-configuration scan, the original policy checker, the cryptographic harness, and the 80-test verification suite, together with the full scan traces, are not part of this submission package. The numbers reported from them were produced by those scripts and are reproduced in the paper from their saved outputs; the included budget and handover summaries record the input hashes of those outputs.

The main scan's TTL grid is $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14, 15, 16, 19, 20, 21, 29, 30, 31\}$; budgets are $\{0, 1, 2, 3, 5, 10, 15, 20, 30\}$. A strict cache reuses when age $< \tau$ and an inclusive cache when age $\leq \tau$; zero always reevaluates, including within the same tick. Hard grant intervals are half-open under both conventions. All gate comparisons include the allow result, decision revision, response reason, and work counters.

A.2 Query Inventory and Complete Histories

Table 10 lists all 26 scheduled labels. F is the former representative/support actor, N the newly assigned actor, Y the young subject S, and C the administrative actor. Unknown actors and unsupported combinations are added by the random checkers. New-schema histories, assumptions, decisions, and frames are in the included `extension_results/fixtures.json`; the original histories' event actions and manual frames belong to the main-scan materials not included in this package. The fixtures specify the complete test contract, including independent grants, temporal boundaries, and superseding snapshots.

The two original authority histories differ before confirmation: one preserves inherited sharing and the other narrows it; both later confirm and then withdraw sharing. The no-contact pair differs between receipt at 7 and review at 19. These original histories are separate from running_S; their 73-versus-zero result is not a measurement of S's combined history. New-schema rights-request receipt and adult-review events can change recorded facts without changing an ordinary query outcome.

Table 10. Scheduled query inventory. Actor and subject identity are bound by the test harness; purpose labels describe supported workflows.

Label	Field	Operation	Purpose
F_PHONE_SUPPORT	phone	read	support
F_ADDRESS_SUPPORT	address	read	support
F_EMAIL_EVENT	email	read	event
F_EMAIL_COPY	email	copy	event
F_EMAIL_PRINT	email	print	event
F_PHONE_COPY	phone	copy	support
F_PHONE_PRINT	phone	print	support
F_PHONE_EVENT	phone	read	event
N_PHONE_SUPPORT	phone	read	support
N_ADDRESS_SUPPORT	address	read	support
N_EMAIL_OUTREACH	email	read	outreach
N_EMAIL_COPY	email	copy	outreach
N_PHONE_DIRECTORY	phone	read	directory
N_EMAIL_PRINT	email	print	event
Y_PHONE_READ	phone	read	self
Y_EMAIL_READ	email	read	self
Y_ADDRESS_READ	address	read	self
Y_PHONE_MANAGE	phone	manage	visibility
Y_EMAIL_MANAGE	email	manage	visibility
F_PHONE_MANAGE	phone	manage	visibility
F_EMAIL_MANAGE	email	manage	visibility
C_EMAIL_ADMIN	email	read	admin
C_PHONE_ADMIN	phone	read	admin
F_EMAIL_ADMIN	email	read	admin
Y_REQUEST	record	request	rights
N_REQUEST	record	request	rights

A.3 Evidence Record Excerpt

The following display condenses the relationship-end record. The interval is half-open; in the machine-readable file, `interval_half_open: [31, 37]` is an array of endpoints interpreted by the schema. Source hashes and detailed residual entries are omitted.

```
history: running_S
event_id: support_relationship_ended
configuration: guard_push; seed: 0; outages: false
target_revision: 5
interval: {start_tick: 31, end_tick: 37, end_exclusive: true}
changed_query_labels: 4; observed_query_labels: 3
unobserved_query_labels: [N_ADDRESS_SUPPORT]
observed_responses: 14; post_budget_observed_responses: 9
false_grants: 0; false_denials: 0; late_false_grants: 0
last_observed_stale_allow_tick: null
applied_revision_of_all_services: unknown
linked_material_observations: 5
possible_recipient_notification: not executed
recipient_deletion: unverified
```

This human-readable display is not literal JSON. The requirement-level sidecar (Section 6.5) is distinct from these revision-bounded records.

A.4 Tests and Interpretation

A relation interpreter checked 2,172,480 original-policy decisions, including 12,480 manually enumerated points, and detected seven specified wrong implementations. A separate cache interpreter matched every scan response and work counter. The retention schema passed 160,000 comparisons over 200 randomized states and detected controls that ignored retention or treated it as a grant. These are shared-semantics checks, not independent legal validation.

The 80 unit tests consist of 29 original experiment tests, 14 relation-interpreter tests, 17 TTL tests, 13 youth/retention tests, and seven evidence/control tests. The seven original wrong controls omit restrictions, ignore dependencies, include an excluded expiry endpoint, allow a grant early, equate reading with management, treat retention as a grant, or disable the rights route with account closure. Boundary tests check exact error-tick sets rather than aggregate counts. Random differential tests are finite and mostly include denied queries; agreement establishes neither legal correctness nor exhaustive system behavior. Eight further validator controls test the evidence semantics, and four further tests cover the hidden-path replay (Appendix A.8); both are reported separately from the 80 tests.

A.5 Counting and Saved-Output Postprocessing

TTL 30 has 27,726 reference-denied and 30,994 reference-allowed queries. Its 3,068 FG include 366 management operations, 126 first-observed actor/field/episode versions, and 2,576 repeated versions; first-observed means first in that modeled trace. The maximum sampled FG offset is 22 ticks after the corresponding flip; Equation 4, not that observation, bounds the idealized model.

The budget table reads saved configuration totals for $b \in \{0, 1, 2, 5, 10\}$ and strict $\tau = b + 1$; reduction is relative to 49,264 Fresh full evaluations and excludes revision checks. The matched handover comparison filters guard-with-push, uninterrupted lookup, five seeds, and $[7, 19)$, pairing by seed, tick, query, and within-tick occurrence to preserve repeated probes. Its independent service

group is F_PHONE_SUPPORT, F_ADDRESS_SUPPORT, and F_EMAIL_EVENT. The CSVs retain per-seed counts, parameters, and input hashes; this aggregation creates no new episodes.

Table 11. Per-history FG/FD. Each cell covers five seeds in each lookup condition (ten episodes).

History	Fresh	TTL 3	TTL 30	Guard	Push only
Authority: keep	0/212	7/148	142/345	0/212	18/52
Authority: confirm	0/209	7/146	166/345	0/209	22/57
Appointment ends	0/135	28/85	344/2	0/135	48/9
Relationship transfer	0/172	28/129	344/342	0/172	48/52
Contact: interim	0/141	26/93	284/2	0/141	38/6
Contact: review	0/159	18/105	138/2	0/159	44/6
Departure / retention	0/9	82/6	1,356/0	0/9	176/0
Superseding changes	0/135	58/111	294/2	0/135	108/9

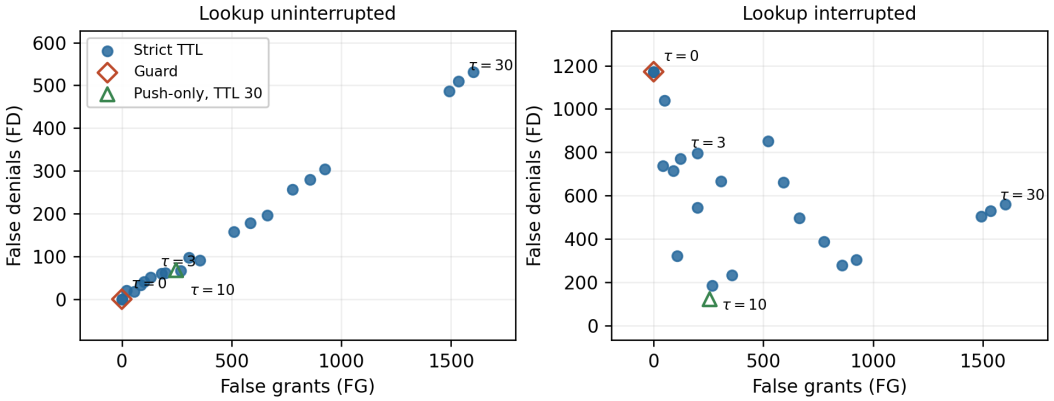


Fig. 2. Discrete FG–FD outcomes for all strict TTL values, the guard, and the strict push-only control, separated by lookup condition. Each panel covers 29,360 decisions per configuration. Points are not connected: parameter alignment can make outcomes nonmonotonic. The guard overlaps the fresh outcome.

A.6 Observed Budget Cells

Figure 3 retains the selected discrete cells, separating finite observed success from the sufficient analytical region.

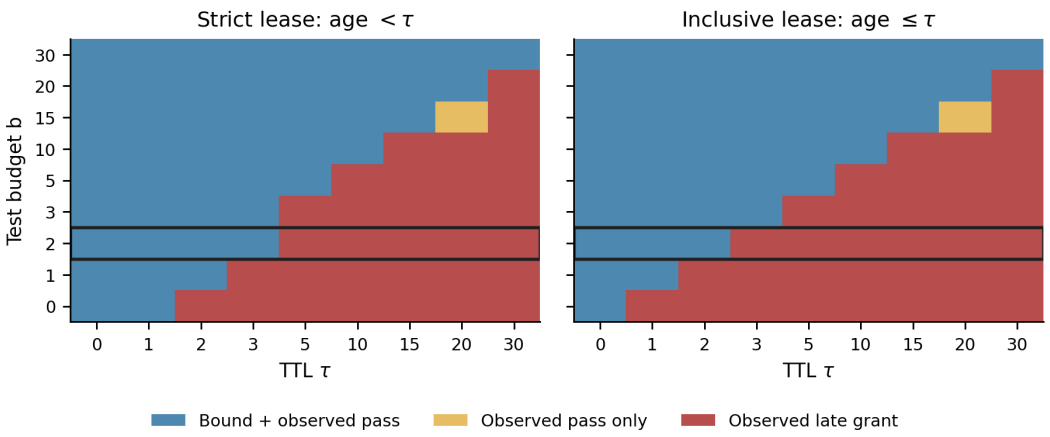


Fig. 3. Selected (τ, b) cells from the full fixed grid. Passing means no late FG in either lookup condition. Blue also satisfies the sufficient main-order bound; gold passes only in the finite observations; red has a late FG. The outlined row is $b = 2$. Axes enumerate discrete, unevenly spaced parameters.

A.7 Additional Saved-Trace Analyses

The revision analysis reads saved configuration totals, request traces, fixtures, commits, and generated records; its manifest records input hashes and parameters. The evidence validator receives a scope contract before inspecting observations. Challenges broaden query or path coverage, end an interval before its deadline, or stipulate an applicable but unfinished notification action, and a complete bounded control is included. The validator reports sampled errors and scope/action gaps separately. The continuity analysis follows the contact restriction from tick 37 through the history's end, keeping a stable requirement identifier, its effective time and deadline 39, its covered revisions, and any explicit superseding decision, and tests that the declared denied query stays denied across covered revisions. All intervals are half-open. Neither the authorization scan nor the cryptographic experiments is rerun for these analyses. The challenge validator is separate from the policy evaluator but belongs to the same research process; it does not authenticate inputs or determine which legal duty applies.

A.8 Hidden-Path Replay

The replay is a separate standard-library program (`hidden_path/`) and is not part of the 46-configuration scan or its counts. A single synthetic promotional-email authorization is revoked by the authoritative policy at commit tick 7; the reference decision is fixed by that event time rather than computed by a gate. An independently supplied deployment inventory names a primary path, which runs the revision guard, and a secondary path, which runs the same TTL gate as the main study with strict $\tau = 30$, no revision check, and no push invalidation. Both paths are filled one tick before the commit. The observation interval is $[7, 14)$ with $b = 2$. Four executions share the event schedule: primary-only probes under a primary-only contract; primary-only probes under the inventory contract; paired probes of both paths, with the secondary path probed every tick

from the deadline; and the same paired probes with a guarded secondary path. The program asserts that adding secondary probes leaves the primary-path responses unchanged and fails, rather than exporting a result, if a changed parameter removes the intended contrast (for example, $\tau \leq b + 1$). Four semantic tests passed. The reported run used Python 3.13.16; the program is deterministic. Outputs include raw executions, contracts, evidence records, a CSV summary, and source hashes.

A.9 Human-Rights Context

ECHR Articles 8, 9, and 11 supply interpretive context rather than directory permissions. HRA 1998 section 6 concerns public authorities; Protocol 1 Article 2 concerns State education and parental convictions, not general parental directory access [12, 13]. Organizational autonomy does not identify who determines processing purposes and means.